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Wang

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(54) **MULTI-PUMP INTEGRATED MIXED FLUID DELIVERY DEVICE**

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E21B 49/08	(2006.01)
F04B 9/02	(2006.01)
F04B 17/03	(2006.01)
F04B 47/06	(2006.01)
F04B 53/16	(2006.01)
F04B 53/22	(2006.01)

(52) **U.S. Cl.**

CPC ***F04B 23/06*** (2013.01); ***E21B 49/081*** (2013.01); ***E21B 49/084*** (2013.01); ***F04B 9/02*** (2013.01); ***F04B 17/03*** (2013.01); ***F04B 47/06*** (2013.01); ***F04B 53/16*** (2013.01); ***F04B 53/22*** (2013.01)

(58) **Field of Classification Search**

CPC .. F04B 23/06; F04B 9/02; F04B 17/03; F04B 47/06; F04B 53/16; F04B 53/22; E21B 49/081; E21B 49/084

See application file for complete search history.

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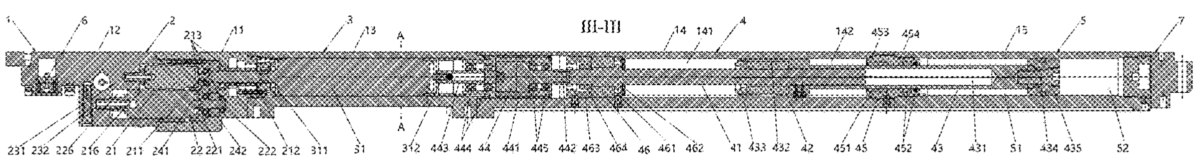
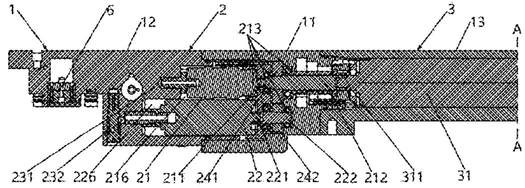
Primary Examiner — Bryan M Lettman

(57) **ABSTRACT**

A multi-pump integrated mixed fluid delivery device includes a hollow cylindrical casing, a hydraulic drive part, a power part, a sample part and a pipeline unit. The hydraulic drive part, the power part, the sample part and the pipeline unit are provided within the casing in sequence. The hydraulic drive part includes multiple hydraulic plunger pumps. The power part includes a dual-shaft motor, wherein an output shaft of the dual-shaft motor is connected with the hydraulic plunger pumps. The sample part includes a lead screw, a screw nut, a piston rod and a sample chamber, wherein one end of the lead screw is connected with another output shaft of the dual-shaft motor, the screw nut is sleeved on the lead screw, the piston rod is fixedly connected with the screw nut. The sample chamber is configured to accommodate a sample.

17 Claims, 6 Drawing Sheets





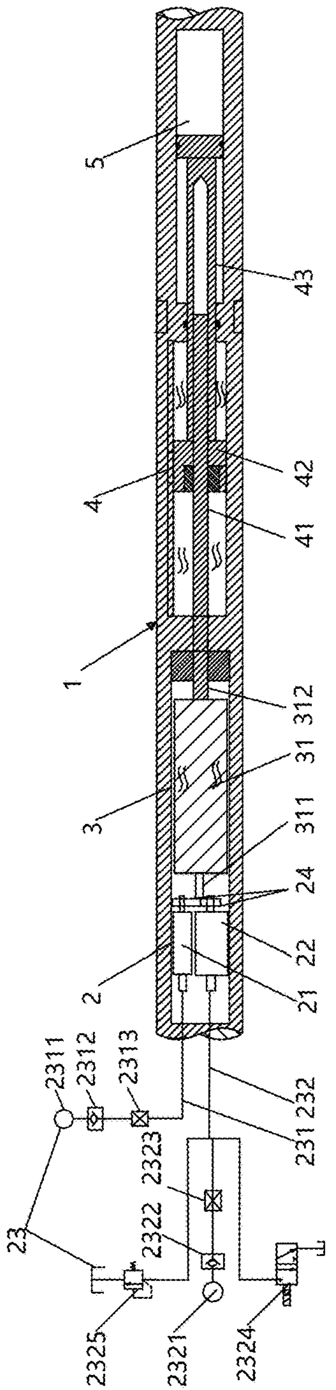

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Fig. 1

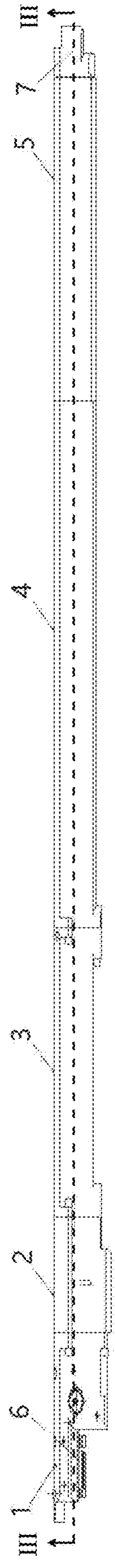
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Fig. 2

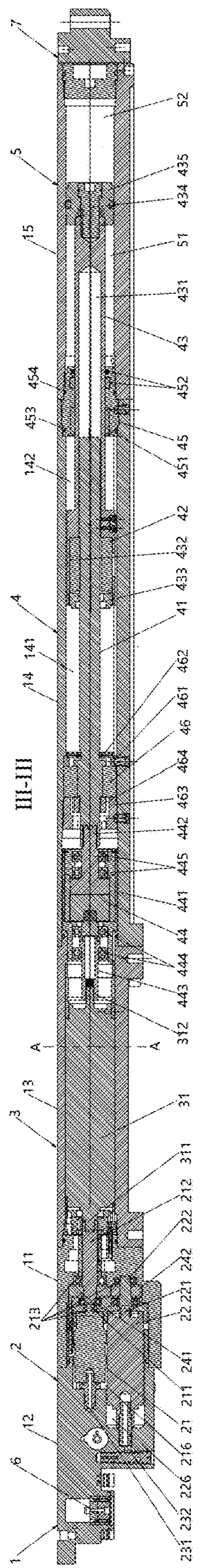
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Fig. 3

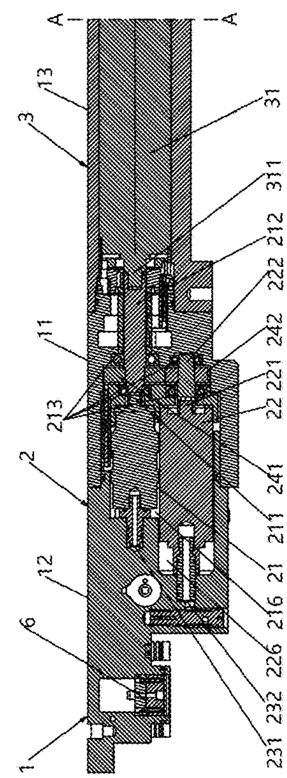
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Fig. 3A

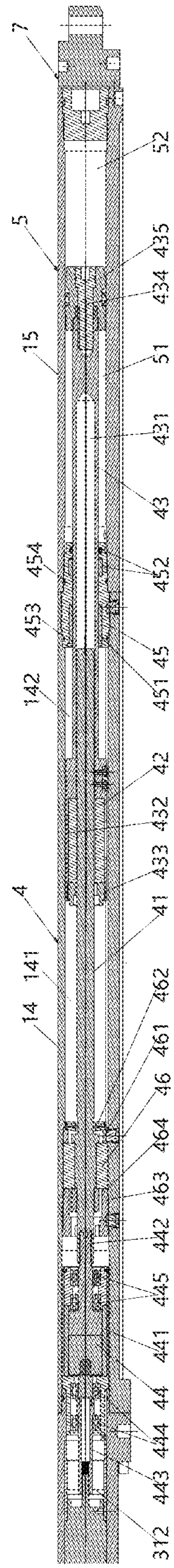
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Fig. 3B

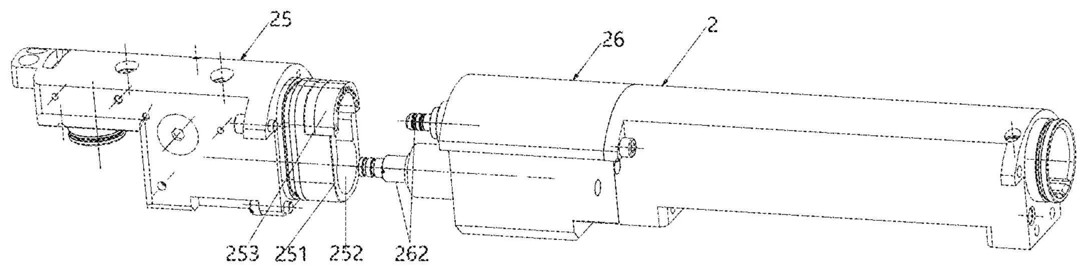

Fig. 4

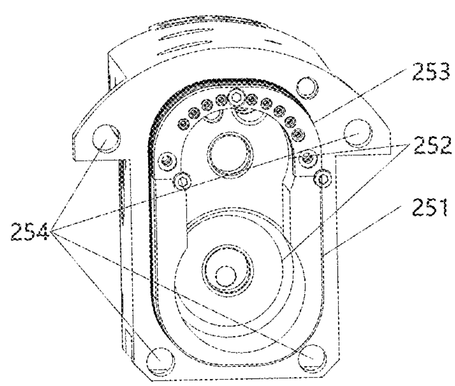

Fig. 5

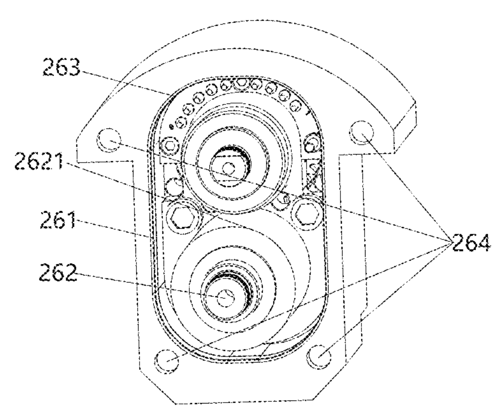

Fig. 6

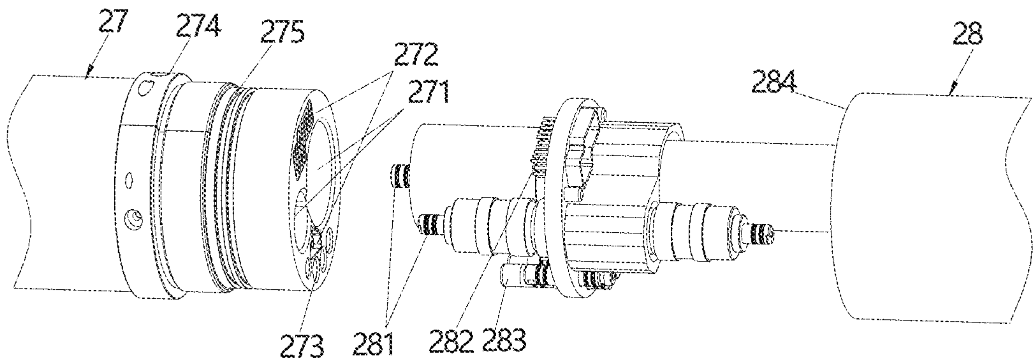

Fig. 7

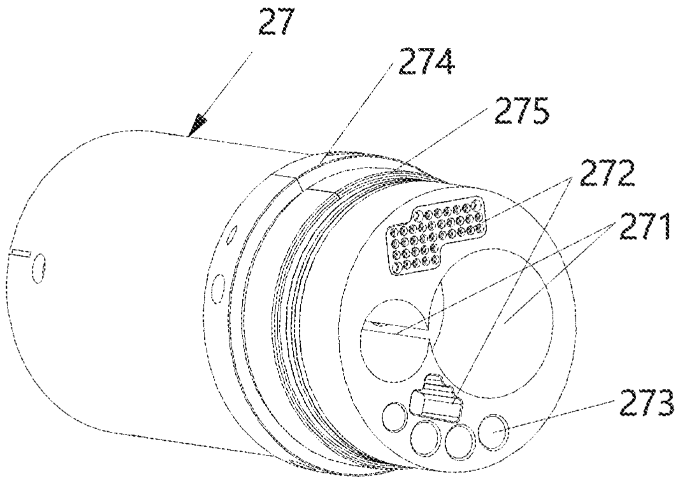

Fig. 8

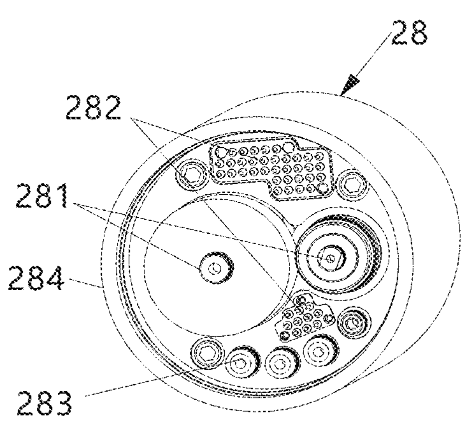

Fig. 9

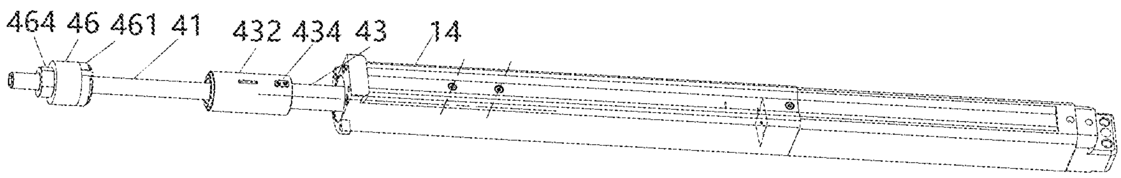

Fig. 10

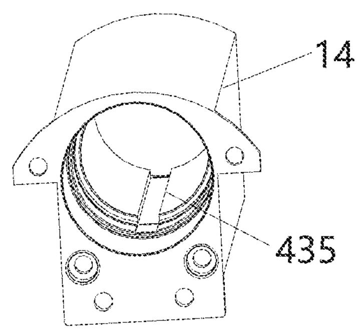

Fig. 11

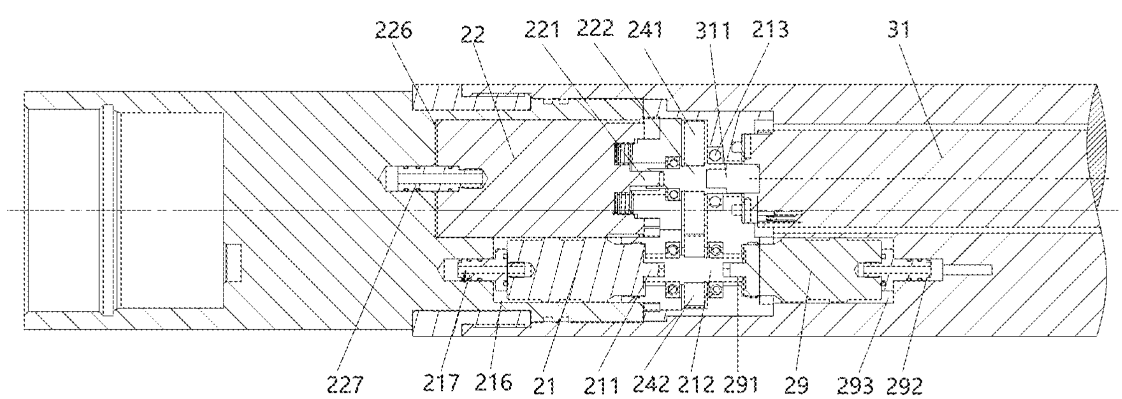

Fig. 12

MULTI-PUMP INTEGRATED MIXED FLUID DELIVERY DEVICE

CROSS REFERENCE OF RELATED APPLICATION

The present invention claims priority under 35 U.S.C. 119(a-d) to CN 202210022910.4, filed Jan. 10, 2022.

BACKGROUND OF THE PRESENT INVENTION

Field of Invention

The present invention relates to the field of downhole petroleum measurement, and more particularly to a multi-pump integrated mixed fluid delivery device, which is able to simultaneously drive multiple pumps through a motor to output high-pressure hydraulic oil with different flow rates.

Description of Related Arts

With the development of the petroleum industry, petroleum exploration and exploration technologies are constantly updated. Formation sampling instruments have always been an important part of exploration equipment in the field of petroleum exploration. They are used to measure various data of current drilling, such as inclination and sampling.

The existing eccentering device itself requires the high-pressure hydraulic oil to be driven, such as the extension of the eccentering device. In addition, some measuring instruments installed on the eccentering device also need the high-pressure hydraulic oil to provide power. Sample cartridge for underground mud.

Generally, the existing eccentering device is only able to provide the high-pressure hydraulic oil with one flow rate (pressure), which is unable to meet the different requirements of different devices, and easily leads to power waste or shortage.

SUMMARY OF THE PRESENT INVENTION

An object of the present invention is to provide a multi-pump integrated mixed fluid delivery device, which is able to simultaneously drive multiple pumps through a motor to output high-pressure hydraulic oil with different flow rates.

Specifically, the present invention provides a multi-pump integrated mixed fluid delivery device, which comprises a hollow cylindrical casing, a hydraulic drive part, a power part, a sample part and a pipeline unit, wherein:

- the hydraulic drive part, the power part, the sample part and the pipeline unit are provided within the casing in sequence;
- the hydraulic drive part comprises multiple hydraulic plunger pumps, wherein an output end of each of the hydraulic plunger pumps is connected with a high-pressure oil output pipe, drive shafts of the plunger pumps are connected with each other through gears which are engaged with each other;
- the power part comprises a dual-shaft motor, wherein an output shaft of the dual-shaft motor, which is provided at one end of the dual-shaft motor, is connected with one of the gears;
- the sample part comprises a lead screw, a screw nut, a piston rod and a sample chamber, wherein one end of the lead screw is connected with another output shaft of the dual-shaft motor, which is provided at another end of the dual-shaft motor, the screw nut is sleeved on the lead screw, the piston rod is fixedly connected with the screw nut, and the sample chamber is configured to accommodate a sample;
- the pipeline unit comprises a cable passage and a hydraulic pipe, wherein:
  - the cable passage is configured to transmit signals, the hydraulic pipe is configured to convey hydraulic oil;
  - the pipeline unit is provided within the casing;
  - the output shaft of the dual-shaft motor drives the plunger pumps to synchronously rotate through the gears, so as to suck hydraulic oil in a mounting chamber of the each of the plunger pumps for pressurization, and then different flow rates of high-pressure hydraulic oil is outputted through the high-pressure oil output pipes respectively; the another output shaft of the dual-shaft motor drives the lead screw to rotate, so that the screw nut drives the piston rod to axially move, so as to suck an external sample into the sample chamber and deliver to a sample cylinder.

The delivery device provided by the present invention is able to simultaneously drive two plunger pumps and one piston pump to work through a dual-shaft motor, and drive three power sources. Moreover, it is able to output different flow rates of high-pressure hydraulic oil by adjusting the power of the two plunger pumps, which expands the application scope of the power. Through the gear linkage, one output shaft of the dual-shaft motor is able to drive two plunger pumps to work at the same time, which simplifies the overall connection structure and meets the space requirements for the casing to adapt to different diameters in the well. The present invention provides a variety of power options for main bases with different needs, and is convenient for installation and maintenance, which greatly improves the work efficiency.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a layout diagram of a multi-pump integrated mixed fluid delivery device provided by the present invention.

FIG. **2** is an overall schematic diagram of the multi-pump integrated mixed fluid delivery device provided by the present invention.

FIG. **3** is a sectional view of FIG. **2** along a central axis of a cylindrical casing.

FIG. **3**A is a partial view of FIG. **3**.

FIG. **3**B is another partial view of FIG. **3**.

FIG. **4** is a schematic diagram of a plugging manner of a plunger sub-casing of the multi-pump integrated mixed fluid delivery device provided by the present invention.

FIG. **5** is a structurally schematic diagram of a first oil pipe module shown in FIG. **4**.

FIG. **6** is a structurally schematic diagram of a first pump module shown in FIG. **4**.

FIG. **7** is a schematic diagram of another plugging manner of the plunger sub-casing of the multi-pump integrated mixed fluid delivery device provided by the present invention.

FIG. **8** is a structurally schematic diagram of a second oil pipe module shown in FIG. **7**.

FIG. **9** is a structurally schematic diagram of a second pump module shown in FIG. **7**.

FIG. **10** is a structurally schematic diagram of a positioning key.

FIG. **11** is a structurally schematic diagram of a feather keyway which matches the positioning key shown in FIG. **10**.

FIG. **12** shows a connection relationship among three pumps.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

The specific structure and implementation process of the technical solution of the present invention will be described in detail with specific embodiments and accompanying drawings as follows. The present invention is able to be applied to the downhole measuring instrument, in which the "main base" mentioned therein refers to the body on which the measuring instrument is installed. "Left" refers to the left side of the diagram while anyone faces the drawings, "right" refers to the right side of the diagram while anyone faces the drawings, and "outside world" refers to the surrounding environment of the downhole measurement location.

As shown in FIG. **1**, a multi-pump integrated mixed fluid delivery device according to a preferred embodiment of the present invention is illustrated, which comprises a hollow cylindrical casing **1**, a hydraulic drive part **2**, a power part **3**, a sample part **4** and a pipeline unit, wherein the hydraulic drive part **2**, the power part **3**, the sample part **4** and the pipeline unit are provided within the casing **1** in sequence.

The hydraulic drive part **2**, which is configured to deliver high-pressure hydraulic oil to an outside world, comprises a first hydraulic plunger pump **21** and a second hydraulic plunger pump **22**, wherein an output end of the first hydraulic plunger pump **21** and an output end of the second hydraulic plunger pump **22** are connected with a first high-pressure oil output pipe **231** and a second high-pressure oil output pipe **232** respectively, a first drive shaft of the first hydraulic plunger pump **21** and a second drive shaft of the second hydraulic plunger pump **22** are connected with a first gear **241** and a second gear **242** respectively, the first gear **241** and the second gear **242** are engaged with each other.

The first hydraulic plunger pump **21** and the second hydraulic plunger pump **22** are able to be same in power or different in power. According to the preferred embodiment of the present invention, a flow rate of the second hydraulic plunger pump **22** is larger than that of the first hydraulic plunger pump **21**. The first and second high-pressure oil output pipes **231**, **232** are configured to provide power for extension and contraction of a push arm of a main base.

The power part **3** comprises a dual-shaft motor **31**, wherein a first output shaft **311** and a second output shaft **312** are provided at two ends of the dual-shaft motor **31** respectively. The first output shaft **311** is connected with the first gear **241** or the second gear **242**, such that when the dual-shaft motor **31** rotates, the first output shaft **311** is able to drive the first gear **241** or the second gear **242** which is engaged with the first output shaft **311** to rotate, for correspondingly driving the second gear **242** or the first gear **241** which is engaged with the first gear **241** or the second gear **242** to rotate, so as to further drive the first drive shaft of the first hydraulic plunger pump **21** and the second drive shaft of the second hydraulic plunger pump **22** to rotate, so that the first hydraulic plunger pump **21** and the second hydraulic plunger pump **22** are able to deliver high-pressure oil with different flow rates to the first and second high-pressure oil output pipes **231**, **232** respectively.

The sample part **4** comprises a lead screw **41**, wherein one end of the lead screw **41** is connected with the second output shaft **312** of the dual-shaft motor **31**, a screw nut **42** is sleeved on the lead screw **41**, a piston rod **43** is fixedly connected with the screw nut **42**, a sample chamber **5** is provided at a tail portion of the sample part **4** for accommodating a sample.

The second output shaft **312** of the dual-shaft motor **31**, the lead screw **41** and the piston rod **43** are axially connected with each other in sequence. When the lead screw **41** rotates, the screw nut **42** moves along an axial direction of the lead screw **41** for driving the piston rod **43** to make a reciprocating movement within the sample chamber **5**.

The pipeline unit comprises a cable passage and a hydraulic pipe, wherein the cable passage is configured to transmit electrical signals on the main base to the dual-shaft motor **31**, the hydraulic pipe is configured to provide various components with the hydraulic oil; the cable passage and the hydraulic pipe are provided within the casing **1**.

When the delivery device provided by the present invention is fixedly connected with the main base, each interface of the delivery device is communicated with a corresponding pipeline on the main base for receiving the electrical signals and the hydraulic oil from the main base. Moreover, the first and second high-pressure oil output pipes **231**, **232** are interconnected with corresponding pipelines of the main base respectively to provide the high-pressure oil with a predetermined pressure for the main base, so as to extend and contract the push arm or drive other components.

Before use, the casing **1** is firstly inserted into a corresponding position of the main base and is fixed, each interface on the casing **1** is interconnected with the corresponding interface on the main base.

During operation, the dual-shaft motor **31** starts to rotate after receiving a signal from the main base, the first output shaft **311** drives the first gear **241** or the second gear **242** to rotate, so as to simultaneously drive the first hydraulic plunger pump **21** and the second hydraulic plunger pump **22** to rotate; after rotation, the first hydraulic plunger pump **21** and the second hydraulic plunger pump **22** absorb the hydraulic oil in the installation chamber, pressurize and then output the pressurized hydraulic oil through the first and second high-pressure oil output pipes **231**, **232** respectively; two output ports are provided on the first and second high-pressure oil output pipes **231**, **232** respectively and are controlled to be opened or closed according to the flow requirement of the main base.

At the same time, the second output shaft **312** of the dual-shaft motor **31** drives the lead screw **41** to rotate for driving the screw nut **42** to move axially, so as to drive the piston rod **43** to move axially. The sample chamber **5** has a sample access hole (not shown in the drawings) for interconnecting with a sample cylinder for storing an external sample and the outside world. When the piston rod **43** moves within the sample chamber **5**, the external sample is sucked into the sample chamber **5**; and when the piston rod **43** does an opposite action, the sample in the sample chamber **5** is pressed into the sample cylinder.

According to the preferred embodiment of the present invention, the dual-shaft motor **31** is able to simultaneously drive two hydraulic plunger pumps and one piston pump to work, and to drive three power sources, and simultaneously the high-pressure hydraulic oil with different flow rates is outputted by adjusting the power of the two hydraulic plunger pumps, which expands the power application range. Through the gear linkage, one output shaft of the dual-shaft motor is able to simultaneously drive two hydraulic plunger pumps to work, which simplifies the overall connection structure and is able to meet the space requirements for the casing to adapt to different diameters in the well. According to the preferred embodiment, the present invention is able to provide a variety of power options for the main base with different needs, and is convenient for installation and maintenance, which greatly improves the work efficiency.

FIG. **2** is an overall schematic diagram of the delivery device provided by the present invention. The first and second high-pressure oil output pipes **231**, **232** are provided within the hydraulic drive part **2**. As shown in FIG. **2**, in order to facilitate the installation and signal transmission of the delivery device, a multi-pin plug **6** is installed at one end (left end) of the casing **1**, and a fixed seat **7** is installed at another end (right end) of the casing **1**, an interface orientation of the multi-pin plug **6** is towards the main base, the multi-pin plug **6** has multiple plug holes; the fixed seat **7** is screwed onto a right end of the sample part **4** through screw threads; the casing **1** has two bolt holes at two ends thereof respectively. During installation, the casing **1** is overall inserted into an installation slot of the main base, an outer surface of the casing **1** is consistent with an outer surface of the main base, the multi-pin plug **6** is plugged into a multi-pin socket on the main base, the fixed seat **7** is inserted into a fixed slot on the main base, and the casing **1** is fixed with the main base through two screw rods which penetrate through the two bolt holes respectively. According to the preferred embodiment of the present invention, the delivery device is convenient for assembly and disassembly, and subsequent maintenance.

Referring to FIG. **3**, in order to facilitate the installation of various components of the delivery device, the casing **1** preferably comprises multiple sub-casings connected with each other in sequence, wherein the multiple sub-casings are a plunger sub-casing **12**, a connection sub-casing **11**, a motor sub-casing **13**, a piston sub-casing **14** and a sample sub-casing **15** respectively. The plunger sub-casing **12** is for installing the first hydraulic plunger pump **21**, the second hydraulic plunger pump **22**, and the first and second high-pressure oil output pipes **231**, **232**. The connection sub-casing **11** is for installing connection components, namely, the first and second gears **241**, **242** which are configured to connect the first and second hydraulic plunger pumps **21**, **22** with the first output shaft **311** of the dual-shaft motor **31** respectively. The motor sub-casing **13** is for installing the dual-shaft motor **31**. The piston sub-casing **14** is for installing the piston part **4**. The sample sub-casing **15** is for defining the sample chamber **5**. The sub-casings are connected with each other in a threaded manner and then are fixed by bolts.

By dividing the entire casing **1** into multiple sub-casings, the components are able to be installed within each corresponding sub-casing in advance, and then connected in a unified manner, which simplifies the installation process of the entire delivery device. In addition, when a component fails, the sub-casing corresponding to the component is able to be disassembled separately for maintenance or replacement. Therefore, the delivery device is improved in maintenance efficiency and reduced in cost.

Referring to FIG. **1**, according to the preferred embodiment of the present invention, a discharge capacity of the first hydraulic plunger pump **21** is smaller than that of the second hydraulic plunger pump **22**, the first and second high-pressure oil output pipes **231**, **232** are independent from each other.

A first filter screen **2313** for filtering impurities, a first one-way stop valve **2312** for avoiding backflow of hydraulic oil, and a first output port **2311** for connecting with an external power are provided on the first high-pressure oil output pipe **231**.

A second filter screen **2323** for filtering impurities, a second one-way stop valve **2322** for avoiding backflow of hydraulic oil, an overflow valve **2325** for oil discharge after exceeding a predetermined pressure, a solenoid valve **2324** for controlling opening and closing of the second high-pressure oil output pipe **232**, and a second output port **2321** for connecting with an external power are provided on the second high-pressure oil output pipe **232**.

The first and second filter screens **2313**, **2323** are able to ensure the cleanliness of the outputted hydraulic oil. The first and second one-way stop valve **2312**, **2322** are able to prevent the hydraulic oil from the first and second output ports **2311**, **2321** from flowing backward into the first and second high-pressure oil output pipes **231**, **232** respectively when the first and second hydraulic plunger pumps do not work. The overflow valve **2325** is able to ensure the stability of the pressure of the second high-pressure oil output pipe **232**, and to drain excess hydraulic oil. While being energized, the solenoid valve **2324** is able to control whether the high-pressure hydraulic oil is outputted to the outside world, and at the same time, to return the high-pressure hydraulic oil discharged from the second hydraulic plunger pump **22** to an oil return pipeline.

As shown in FIG. **3**, in order to shorten an overall length of the delivery device, the first and second hydraulic plunger pumps **21**, **22** are set within the plunger sub-casing **12** in parallel, the first drive shaft **211** of the first hydraulic plunger pump **21** and a second drive shaft **221** of the second hydraulic plunger pump **22** are provided at a same direction, namely, a right side of the first hydraulic plunger pump **21** and a right side of the second hydraulic plunger pump **22** respectively, or a left side of the first hydraulic plunger pump **21** and a left side of the second hydraulic plunger pump **22** respectively; the first and second gears **241**, **242** are directly installed on the first and second drive shafts **211**, **221** respectively and are engaged with each other.

When the first and second drive shafts **211**, **221** are not long enough, a first transmission shaft and a second transmission shaft for extension are able to be installed between the first and second hydraulic plunger pumps **21**, **22** and the dual-shaft motor **31** respectively; and then the first and second gears **241**, **242** are installed on the first and second transmission shafts respectively. According to the preferred embodiment, the first output shaft **311** of the dual-shaft motor **31** is connected with the first drive shaft **211** of the first hydraulic plunger pump **21** through the first transmission shaft **212**, the first gear **241** is fixedly installed on an external surface of the first transmission shaft **212**.

To correspond to the position of the first gear **241**, the second drive shaft **221** of the second hydraulic plunger pump **22** is connected with the second transmission shaft **222**, the second gear **242** is fixedly installed on an external surface of the second transmission shaft **222**, external teeth of the second gear **242** are engaged with external teeth of the first gear **241**. The first transmission shaft **212** is longer than the second transmission shaft **222**, and a diameter of the first gear **241** is larger than that of the second gear **242**.

Gears with different diameters are able to selected according to installation positions of the first and second hydraulic plunger pumps **21**, **22**, and moreover, the rotating speed of the first and second hydraulic plunger pumps **21**, **22** is able to be controlled through the gears with different diameters. To limit positions of the first and second gears **241**, **242**, four first bearings **213** for limit and support are installed at two sides of the first and second gears **241**, **242**.

While working, the first output shaft **311** of the dual-shaft motor **31** drives the first transmission shaft **212** to rotate, so as to further drive the first drive shaft **211** of the first hydraulic plunger pump **21** to rotate. When the first transmission shaft **212** rotates, the first gear **241** fixed on the first transmission shaft **212** simultaneously drives the second gear **242** which is engaged with the first gear **241** to rotate, so that the second hydraulic plunger pump **22** is indirectly driven to rotate through the second transmission shaft **222** and the second drive shaft **221**, thereby achieving that the dual-shaft motor **31** simultaneously drives the first and second hydraulic plunger pumps **21**, **22** to rotate through the first output shaft **311** of the dual-shaft motor **31**.

Further, the first and second hydraulic plunger pumps **21**, **22** are installed within a first plunger chamber **216** and a second plunger chamber **226** of the plunger sub-casing **12** through external threads respectively; the first and second plunger chambers **216**, **226**, which are provided in parallel and interconnected with each other, are full of hydraulic oil. When the first and second hydraulic plunger pumps **21**, **22** work, the normal-pressure hydraulic oil in the first and second plunger chambers **216**, **226** is directly sucked into the first and second hydraulic plunger pumps **21**, **22** respectively, and then is pressurized, and then the pressurized hydraulic oil is outputted through the first and second high-pressure oil output pipes **231**, **232** respectively. The first and second plunger chambers **216**, **226** are interconnected with the oil return pipeline of the delivery device for recycling the hydraulic oil. Moreover, in spite that the first and second plunger chambers **216**, **226** are interconnected with each other, the first and second high-pressure oil output pipes **231**, **232**, which are connected with the first and second plunger chambers **216**, **226** respectively, are independent from each other.

As shown in FIGS. **4-6**, a connection manner between the first and second hydraulic plunger pumps and the first and second high-pressure oil output pipes is illustrated. In the plunger sub-casing **12**, the first and second hydraulic plunger pumps **21**, **22** are connected with the first and second high-pressure oil output pipes **231**, **232** respectively through a plug-in structure. The plug-in structure comprises a first oil pipe module **25** with the first and second high-pressure oil output pipes **231**, **232**, and a first pump module **26** with the first and second hydraulic plunger pumps **21**, **22**.

A rectangular convex structure **251**, which is provided at a connection end of the first oil pipe module **25**, has two first stepped insertion holes **252** which are interconnected with the first and second high-pressure oil output pipes **231**, **232** respectively. A distance is provided between an external side of the rectangular convex structure **251** and an edge of each of the first stepped insertion holes **252**, in such a manner that the rectangular convex structure **251** has a corresponding width. A first multi-pin socket **253**, which is fixed on an upper portion of the rectangular convex structure **251** through studs. Multiple socket holes are evenly distributed in a semi-arc shape based on a shape of the upper portion of the rectangular convex structure **251**.

A rectangular concave structure **261** is provided at an output end of the first pump module **26** (which is corresponding to the first oil pipe module **25**). An inner diameter of the rectangular concave structure **261** is equal to an outer diameter of the rectangular convex structure **251**, such that the rectangular convex structure **251** is fully inserted into the rectangular concave structure **261**. A first multi-pin plug **263** which matches the first multi-pin socket **253** is installed on the rectangular concave structure **261**. Two oil pipe joints **262** are provided within the rectangular concave structure **261** and match the first stepped insertion holes **252** respectively. Two cylindrical fixed columns **2621** are sleeved on the oil pipe joints **262** respectively, so that after the oil pipe joints **262** are inserted into the first stepped insertion holes **252**, the fixed columns **2621** are engaged with the first stepped insertion holes **252** for sealing, so as to prevent the oil pipe joints **262** from radial movement. Furthermore, the fixed columns **2621** are also able to play the role of positioning to avoid the relative rotation of the first oil pipe module **25** and the first pump module **26** after insertion.

The first oil pipe module **25** has multiple first bolt holes **254** at a periphery thereof. The first pump module **26** has multiple second bolt holes **264**, which match the first bolt holes **254** respectively at a periphery thereof. When the first oil pipe module **25** and the first pump module **26** are plugged with each other, the rectangular convex structure **251** is engaged with the rectangular concave structure **261**, the oil pipe joints **262** are inserted into the first stepped insertion holes **252** respectively, multiple pins of the first multi-pin plug **263** are inserted into the multiple socket holes of the first multi-pin socket **253** respectively, and then bolts pass through the first bolt holes **254** and the second bolt holes **264** which are aligned after insertion to fix the first oil pipe module **25** with the first pump module **26**; and then the first oil pipe module **25** and the first pump module **26** are fixed within the plunger sub-casing **12**, or directly form the plunger sub-casing **12**.

According to the preferred embodiment of the present invention, the first multi-pin socket **253** and the first multi-pin plug **263** may be an independent module respectively; the rectangular convex structure **251** and the rectangular concave frame **261** have two openings with the same shape respectively, and the first multi-pin socket **253** and the first multi-pin plug **263** are fixed in the two openings by bolts respectively.

The plug-in structure of this embodiment is suitable for the case where only one measuring instrument is installed on the main base, that is, only one measuring instrument is installed on the main base, or the delivery device is located at the tail of the main base, and there is no need to provide functions other than high-pressure oil and sample collection for other measuring instrument.

As shown in FIGS. **7** to **8**, another connection manner between the first and second hydraulic plunger pumps, and the first and second high-pressure oil output pipes is illustrated. The first and second hydraulic plunger pumps **21**, **22** are connected with the first and second high-pressure oil output pipes **231**, **232** respectively through a screw structure. The screw structure comprises a second oil pipe module **27** with the first and second high-pressure oil output oil pipes **231**, **232**, and a second pump module **28** with the first and second hydraulic plunger pumps **21**, **22**. The second oil pipe module **27** and the second pump module **28** are cylindrical. Two second stepped insertion holes **271**, a second multi-pin socket **272** for signal output, and multiple delivery ports **273** for delivering the hydraulic oil or the sample are provided at a connection end of the second oil pipe module **27**.

Preferably, the two stepped insertion holes **271** are interconnected with the first and second high-pressure oil output pipes **231**, **232** respectively. The second multi-pin socket **272** comprises two multi-pin sub-sockets which are installed on the connection end of the second oil pipe module **27** through slots on the second oil pipe module **27**. The multiple delivery ports **273** are interconnected with pipelines passing through the exterior of the second oil pipe module **27** respectively. The pipelines here refer to the passages located inside the delivery device to delivery hydraulic oil or samples for another connected device, and are not passages serving or serving only the delivery device provided by the present invention.

Two stepped insertion columns **281** for matching the two second stepped insertion holes **271**, a second multi-pin plug **282** for matching the second multi-pin socket **272**, and multiple delivery plugs **283** for matching the multiple delivery ports **273** respectively are provided at an output end of the second pump module **28**.

A threaded sleeve **274** is provided at an external surface of the connection end of the second oil pipe module **27** for fixedly connecting the second oil pipe module **27** with the second pump module **28** after insertion.

During installation, the second oil pipe module **27** and the second pump module **28** are plugged with each other in an end-to-end manner, the two stepped insertion columns **281** are plugged into the two second stepped insertion holes **271** respectively, the two second multi-pin plugs **282** are plugged into the second multi-pin socket **272**, and the multiple delivery plugs **283** are plugged into the multiple delivery ports **273** respectively.

Multiple first sealing rings **275** are provided at the external surface of the connection end of the second oil pipe module **27**, an annular protrusion **284** is provided at an outer circumference of a connection end of the second pump module **28**, an external surface of the annular protrusion **284** has multiple external threads. After the second oil pipe module **27** and the second pump module **28** are plugged with each other, an internal surface of the annular protrusion **284** is sleeved on the external surface of the connection end of the second oil pipe module **27**, and are in sealing contact with the multiple first sealing rings **275**. The second oil pipe module **27** and the second pump module **28** are connected with each other by the threaded sleeve **274** through the external threads of the annular protrusion **284**, and then are fixed with each other by bolts.

According to the preferred embodiment of the present invention, the stepped insertion columns **281** and the multiple delivery plugs **283** are capable of positioning after installation, so as to prevent the second oil delivery module **27** and the second pump module **28** which are connected with each other from relatively rotating. The multiple delivery plugs **283** and the multiple delivery ports **273** form a delivery channel for interconnecting with another device, so as to provide corresponding hydraulic oil or sample for the another device.

The screw structure provided by the present invention is suitable for installing multiple measuring instruments on the main base at the same time. Through the delivery channel, the multiple measuring instruments are able to share a set of hydraulic oil output pipes so that it is not necessary to install a corresponding transmission channel for every measuring instrument, which greatly reduces the diameter of the entire main base and simplifies the design of the main base.

Through the segmented installation structure, each component is able to be disassembled independently, which further improves the maintenance efficiency and reduces the overall cost.

Preferably, a reducer **44** is provided between the second output shaft **312** of the dual-shaft motor **31** and the lead screw **41**, the reducer **44** is installed at an end portion of the piston sub-casing **14** through a mounting base **441**, an output shaft **442** of the reducer **44** is connected with an end portion of the lead screw **41**, an external surface of the mounting base **441** is in contact with an internal surface of the piston sub-casing **14**, and the mounting base **441** is fixed with the piston sub-casing **14** through bolts.

In order to facilitate the connection between the second output shaft **312** of the dual-shaft motor **31** and the reducer **44**, a connection shaft **443** for extending a connection distance is provided between the second output shaft **312** and the reducer **44**. The connection shaft **443** extends into the mounting base **441** and is connected with the reducer **44**, two support bearings **444** are installed on one end portion of the connection shaft **443** which extends into the mounting base **441** for supporting and limiting the connection shaft **443**. Similarly, two second bearings **445** for support and limit are installed on the output shaft **442** of the reducer **44**. The reducer **44** is able to adjust a rotating speed of the lead screw **41** and improve a suction efficiency of the piston rod **43**.

Referring to FIG. **3**, preferably, an isolation sleeve **45**, which is provided in a middle portion of the piston rod **43**, is connected with the piston sub-casing **14** and the sample sub-casing **15** through external threads of the isolation sleeve **45** respectively, and then is fixed with the piston sub-casing **14** and the sample sub-casing **15** through bolts which pass through the piston sub-casing **14** and the sample sub-casing **15**. One end of the isolation sleeve **45** close to the screw nut **42** has an inner annular groove. A metal ring **451**, which contacts the external surface of the piston rod **43**, is provided within the inner annular groove through interference fit. A movable sealing ring **452** is provided within an inner hole which is away from the screw nut **42** and in contact with the piston rod **43**. Two sealing grooves **453**, which match the piston sub-casing **14** and the sample sub-casing **15** respectively, are provided on an external surface of the isolation sleeve **45**. Two second sealing rings **454** are provided within the sealing grooves **453** respectively.

According to the preferred embodiment of the present invention, the isolation sleeve **45** not only serves as a connecting piece for connecting the piston sub-casing **14** with the sample sub-casing **15**, but also serves as a spacer between the screw nut **42** and the sample chamber **5**. There is sliding friction between the metal ring **451** and the piston rod **43**. The metal ring **451** is relatively high in hardness, so its own wear loss is able to be prolonged and the piston rod **43** is prevented from directly wearing the isolation sleeve **45**. At the same time, due to the rigid support effect, the metal ring **451** is also able to reduce the wear of the movable sealing ring **452**, so as to reduce the replacement frequency thereof. Multiple movable sealing rings **452** are able to be provided as required. According to the preferred embodiment of the present invention, there are two movable sealing rings **452**.

After being isolated by the isolation sleeve **45**, a hydraulic oil chamber is formed in the piston sub-casing **14** for the movement of the screw nut **42**, the sample chamber **5** is provided within the sample sub-casing **15**, and an oil pressure of the hydraulic oil chamber is normal pressure.

Further, an accommodating passage **431** for allowing the lead screw **41** to enter is provided within the piston rod **43**, the piston rod **43** is sleeved on the screw nut **42** through an opening **432** of the accommodating passage **431**. A lock ring **433** for locking the screw nut **42** within an interior of the accommodating passage **431** is provided at the opening **432**. An external surface of the opening **432** is in contact with the internal surface of the piston sub-casing **14**. A piston **435**, which has a diameter as same as an inner diameter of the piston sub-casing **14**, is fixed at one end of the piston rod **43** away from the screw nut **42**. The piston **435** divides the sample chamber **5** into a left chamber **51** and a right chamber **52**, the left chamber **51** and the right chamber **52** are interconnected with the outside world through pipelines respectively, and simultaneously are interconnected with the sample cylinder of the main base.

The accommodating passage **431** of the piston rod **43** is not in contact with the lead screw **41** for avoiding the wear on the threads of the lead screw **41**. The piston **435** is able to be made of metal or flexible materials. An axial through-hole is provided in a middle of the piston **435**. The piston **435** is fixed with an end portion of the piston rod **43** through a screw **434** which passes through the axial through-hole. When the piston **435** is made of metal, there are multiple sealing rings on the external surface of the piston **435** which are in contact with the internal surface of the sample chamber **5**.

While working, the dual-shaft motor **31** drives the lead screw **41**, the screw nut **42** and the piston rod **43** to synchronously rotate. When the piston **435** moves rightwards, the right chamber **52** is squeezed to deliver the sample in the right chamber **52** to the sample cylinder; and at this time, due to the right movement of the piston **435**, a suction force is generated in the left chamber **51** for sucking the external sample (such as mud) through a channel. When the piston **435** moves to the rightmost end of the right chamber **52**, the squeeze signal is fed back to the dual-shaft motor **31** (the forward time and the reverse time of the dual-shaft motor **31** are able to be preset according to the stroke of the piston **435**), the dual-shaft motor **31** begins to rotate reversely for driving the piston **435** to move towards the left chamber **51**; and the working way at this time is opposite to that when the piston moves towards the right chamber **52**. Under the squeeze of the piston **435**, the sample in the left chamber **51** is delivered to the sample cylinder, and the external sample is sucked into the right chamber **52** through a channel. When the piston **435** moves to the leftmost end of the left chamber **51**, the piston **435** reverses again and moves towards the right chamber **52**. The process loops until the sample cylinder completes the collection.

In actual use, the small volume of the sample chamber **5** and the large volume of the sample cylinder, the difference between the two may be 10-30 times. Therefore, multiple forward and reverse cycles of the dual-shaft motor **31** are required to meet the collection needs of the sample cylinder. One-way valves are provided on the channels connecting the left chamber **51** and the right chamber **52** with the outside world and the sample cylinder respectively, so as to prevent the reverse flow of the sample in the left chamber **51** and the right chamber **52** when the piston **435** moves leftwards and rightwards.

Further, a limit bearing **46** is provided at one end of the lead screw **41** which is connected with the reducer **44**, a baffle ring **461** and a disc spring **462** are provided at one side of the limit bearing **46** which is away from the reducer **44**, a fixed sleeve **463** and a lock nut **464** are provided at another side of the limit bearing **46** which is close to the reducer **44**, and the lock nut **464** is screwed on the lead screw **41** for fixing the limit bearing **46**.

The limit bearing **46** is able to limit the lead screw **41** to an axial line thereof, the baffle ring **461** and the fixed sleeve **463** are able to limit the position of the limit bearing **46** for avoiding the axial movement of the limit bearing **46**. A step protruding toward an axial center is provided on the inner surface of the piston sub-casing **14** where the limit bearing **46** is installed. The baffle ring **461** is blocked by the step. After installation, the fixed sleeve **463** is fixed by bolts passing through the piston sub-casing **14**.

While working, the piston rod **41** moves leftwards, the disc spring **462** is squeezed to produce a thrust, the thrust is applied to the opening **432** of the piston rod **43** for preventing the opening **432** of the piston rod **43** from being seized with the lead screw **41**.

According to the preferred embodiment of the present invention, the external surface of the opening **432** of the accommodating passage **431** is as same as an inner diameter of the piston sub-casing **14**. A hydraulic oil chamber body of the piston sub-casing **14** is divided into a left hydraulic oil chamber **141** and a right hydraulic oil chamber **142**.

The hydraulic oil in the left hydraulic oil chamber **141** and the right hydraulic oil chamber **142** is normal-pressure oil, which does not output hydraulic power to the outside world, but is only used to keep the pressure in the hydraulic oil chamber at the same level as the outside pressure, so as to avoid the deformation of the piston sub-casing **14**. The left hydraulic oil chamber **141** is interconnected with the right hydraulic oil chamber **142** through a channel. When the screw nut **42** moves leftwards and rightwards, the hydraulic oil in the left hydraulic oil chamber **141** and the hydraulic oil in the right hydraulic oil chamber **14** circulate with each other.

Referring to FIGS. **10** and **11**, in order to limit the piston rod **43** only to move axially, a convex positioning key **434** is provided on the external surface of the opening **432** of the piston rod **43**. A feather keyway **435** matching the positioning key **434** is provided within the piston sub-casing **14**. The feather keyway **435** extends along an axial direction of the piston sub-casing **14**. A length of the feather keyway **435** is corresponding to a movable distance of the screw nut **42**. The positioning key **434** is fixed on the external surface of the opening **432** through a screw which simultaneously passes through the opening **432** and the screw nut **42**. Moreover, the feather keyway **435** is able to act as a channel for interconnecting the left hydraulic oil chamber **141** and the right hydraulic oil chamber **142**, that is, a width of the feather keyway **435** is larger than a width of the positioning key **434**, so as to leave enough space for hydraulic oil to pass through.

Referring to FIG. **12**, based on the above-mentioned technical solution that a dual-shaft motor drives two hydraulic plunger pumps, another technical solution that a dual-shaft motor drives three hydraulic plunger pumps is proposed. The above-mentioned technical solution and the present technical solution are basically same in structure, the difference is that in the technical solution of the three hydraulic plunger pumps, two hydraulic plunger pumps with low flow rate are provided at two sides of the transmission shaft with low flow rate respectively, which is explained in detail as follows.

The first hydraulic plunger pump **21** and the second hydraulic plunger pump **22** are installed within the plunger sub-casing **12** in parallel, the first output shaft **211** of the first hydraulic plunger pump **21** and the second output shaft of the second hydraulic plunger pump **22** are provided in the same direction, the first gear **241** and the second gear **242** are directly installed on the first drive shaft **211** of the first plunger pump **21** and the second drive shaft **221** of the second plunger pump **22** respectively, and engaged with each other.

The first output shaft **311** of the dual-shaft motor **31** is connected with the second drive shaft **221** of the second plunger pump **22** through the second transmission shaft **222**, and the first gear **241** is fixedly installed on the external surface of the second transmission shaft **222**.

To match the first gear **241**, the first drive shaft **211** of the first hydraulic plunger pump **21** is connected with one end of the first transmission shaft **212**. The second gear **242** is fixedly installed on the external surface of the first transmission shaft **212**. The external teeth of the second gear **242** are engaged with the external teeth of the first gear **241**. Another end of the first transmission shaft **212** is connected with a third drive shaft **291** of the third hydraulic plunger pump **29**.

Four first bearings **213** are provided at two sides of the first gear **241** and the second gear **242** respectively for limit and support.

While working, the first output shaft **311** of the dual-shaft motor **31** drives the second transmission shaft **222** to rotate, so as to further drive the second drive shaft **221** of the second hydraulic plunger pump **22** to rotate. When the second transmission shaft **222** rotates, the first gear **241** fixed on the second transmission shaft **222** synchronously drives the second gear **242** to rotate, so as to indirectly synchronously drive the second hydraulic plunger pump and the third hydraulic plunger pump through the first transmission shaft **212** to work, thereby achieving that the dual-shaft motor **31** drives the first hydraulic plunger pump **21**, the second hydraulic plunger pump **22** and the third hydraulic plunger pump **29** in sequence to simultaneously work. The characteristics of the third hydraulic plunger pump **29** are as same as those of the first and second hydraulic plunger pumps **21**, **22**. The third hydraulic plunger pump **29** outputs the high-pressure hydraulic oil with a certain flow rate. The three hydraulic plunger pumps are able to output the high-pressure hydraulic oil with same flow rate, or the high-pressure hydraulic oil with various flow rates. Each device that needs pressure is able to be connected to the corresponding output port as needed, and the high-pressure oil output pipe that is not used is able to be closed through a valve.

The first hydraulic plunger pump **21**, the second hydraulic plunger pump **22** and the third hydraulic plunger pump **29** are installed within a first plunger chamber **216**, a second plunger chamber **226** and a third plunger chamber **293** of the plunger sub-casing **12** respectively through external threads. The first and second plunger chamber **216**, **226** are provided in parallel and interconnected with each other. The third plunger chamber **293** is opposite to and interconnected with the first plunger chamber **216**. The first, second and third plunger chambers **216**, **226**, **293** are full of hydraulic oil. When the first, second and third hydraulic plunger pumps **21**, **22**, **29** work, the normal-pressure hydraulic oil in the first, second and third plunger chambers **216**, **226**, **293** is sucked thereinto and then pressurized, and then is outputted by the first high-pressure oil output pipe **231**, the second high-pressure oil output pipe **232**, and a third high-pressure oil output pipe **292**. The first, second and third plunger chambers **216**, **226**, **293** are interconnected with the oil return pipeline of the delivery device respectively to realize the recycling of hydraulic oil. Moreover, in spite that the first, second and third plunger chambers **216**, **226**, **293** are interconnected with each other, the first, second and third high-pressure oil output pipes **231**, **232**, **292** are independent from each other.

By now, those skilled in the art will recognize that although various exemplary embodiments of the present invention have been shown and described in detail herein, other variations or modifications consistent with the principles of the present invention may be determined directly or derived from the present disclosure without departing from the spirit and scope of the present invention. Therefore, the scope of the present invention should be understood and deemed to cover all such other variations or modifications.

What is claimed is:

1. A multi-pump integrated mixed fluid delivery device, which comprises a hollow cylindrical casing, a hydraulic drive part, a power part, a sample part and a pipeline unit, wherein:

the hydraulic drive part, the power part, the sample part and the pipeline unit are provided within the casing in sequence;

the hydraulic drive part comprises multiple hydraulic plunger pumps, wherein an output end of each of the plunger pumps is connected with an oil output pipe, drive shafts of the hydraulic plunger pumps are connected with each other through gears which are engaged with each other;

the power part comprises a dual-shaft motor, wherein a first output shaft of the dual-shaft motor, which is provided at one end of the dual-shaft motor, is connected with one of the gears;

the sample part comprises a lead screw, a screw nut, a piston rod and a sample chamber, wherein one end of the lead screw is connected with a second output shaft of the dual-shaft motor, which is provided at another end of the dual-shaft motor, the screw nut is sleeved on the lead screw, the piston rod is fixedly connected with the screw nut, and the sample chamber is configured to accommodate a sample;

the pipeline unit comprises a cable passage and multiple oil output pipes comprising the oil output pipe, wherein:

the cable passage is configured to transmit signals, the oil output pipes are configured to convey hydraulic oil;

the pipeline unit is provided within a mounting base of the hydraulic drive part, the power part and the sample part;

the first output shaft of the dual-shaft motor drives the hydraulic plunger pumps to synchronously rotate through the gears, so as to suck hydraulic oil in a mounting chamber of each of the multiple hydraulic plunger pumps for pressurization, and then different flow rates of pressurized hydraulic oil are outputted through the oil output pipes respectively; the second output shaft of the dual-shaft motor drives the lead screw to rotate, so that the screw nut drives the piston rod to axially move, so as to suck an external sample into the sample chamber and deliver to a sample cylinder.

2. The multi-pump integrated mixed fluid delivery device according to claim **1**, wherein there are two hydraulic plunger pumps which are a first hydraulic plunger pump and a second hydraulic plunger pump, a first oil output pipe and a second oil output pipe are connected with an output end of the first hydraulic plunger pump and an output end of the second hydraulic plunger pump respectively, a first gear of the gears is installed on a first drive shaft of the first hydraulic plunger pump, a second gear of the gears is installed on a second drive shaft of the second hydraulic plunger pump, the first gear of the gears is engaged with the second gear of the gears.

3. The multi-pump integrated mixed fluid delivery device according to claim **1**, wherein a multi-pin plug is installed at one end of the casing, and a fixed seat is installed at another end of the casing, the casing is connected and fixed with a main base connecting the cable passage having cables therein and oil pipes comprising the oil outlet pipes through the multi-pin plug and the fixed seat.

4. The multi-pump integrated mixed fluid delivery device according to claim **2**, wherein:

the casing comprises multiple sub-casings connected with each other in sequence;

the multiple sub-casings are a plunger sub-casing, a connection sub-casing, a motor sub-casing, a piston sub-casing and a sample sub-casing respectively;

the plunger sub-casing is for installing the first hydraulic plunger pump and the second hydraulic plunger pump;

the connection sub-casing is for installing the first and second gears which are configured to connect the first and second hydraulic plunger pumps with the output shaft of the dual-shaft motor respectively;

the motor sub-casing is for installing the dual-shaft motor;

the piston sub-casing is for installing the piston part;

the sample sub-casing is for defining the sample chamber.

5. The multi-pump integrated mixed fluid delivery device according to claim **4**, wherein a discharge capacity of the first hydraulic plunger pump is smaller than that of the second hydraulic plunger pump, the first and second oil output pipes are independent from each other.

6. The multi-pump integrated mixed fluid delivery device according to claim **4**, wherein:

in the plunger sub-casing, the first and second oil output pipe form an independent oil pipe module, the first and second hydraulic plunger pumps form an independent pump module, the oil pipe module is connected with the pump module through a plug-in structure;

the plug-in structure comprises a rectangular convex structure which is provided at a connection end of the oil pipe module, a multi-pin socket is installed on the rectangular convex structure, the rectangular convex structure has two insertion holes which are interconnected with the first and second high-pressure oil output pipes respectively;

a rectangular concave structure is provided at an output end of the pump module, a multi-pin plug which matches the multi-pin socket is installed on the rectangular concave structure, two oil pipe joints are provided within the rectangular concave structure and match the two insertion holes respectively;

the oil pipe module has multiple first bolt holes at a periphery thereof, the pump module has multiple second bolt holes, which match the first bolt holes respectively at a periphery thereof.

7. The multi-pump integrated mixed fluid delivery device according to claim **4**, wherein:

the first and second oil output pipes form an independent oil pipe module, the first and second hydraulic plunger pumps form an independent pump module, the oil pipe module is connected with the pump module through a screw structure;

the screw structure comprises two insertion holes, a multi-pin socket for connecting with cables, and multiple delivery ports for delivering hydraulic oil or samples, wherein the two insertion holes are provided in the oil pipe module and interconnected with the first and second oil output pipes respectively;

the multi-pin socket comprises a first multi-pin sub-socket and a second multi-pin sub-socket;

two insertion columns for matching the two insertion holes, a multi-pin plug for matching the multi-pin socket, and multiple delivery plugs for matching the multiple delivery ports respectively are provided at an output end of the pump module;

a threaded sleeve is provided at an external surface of the connection end of the oil pipe module.

8. The multi-pump integrated mixed fluid delivery device according to claim **4**, wherein the second output shaft of the dual-shaft motor is connected with a reducer through a connection shaft and then is connected with the lead screw, the reducer is installed at an end portion of the piston sub-casing through a mounting base, an output shaft of the reducer is connected with an end portion of the lead screw, the connection shaft extends into the mounting base and is connected with the reducer, two support bearings are installed on one end portion of the connection shaft.

9. The multi-pump integrated mixed fluid delivery device according to claim **8**, wherein a limit bearing is provided at one end of the lead screw which is connected with the reducer, a baffle ring and a disc spring are provided at one side of the limit bearing which is away from the reducer, a fixed sleeve and a lock nut are provided at another side of the limit bearing which is close to the reducer, and the lock nut is screwed on the lead screw for fixing the limit bearing.

10. The multi-pump integrated mixed fluid delivery device according to claim **8**, wherein an accommodating passage for allowing the lead screw to enter is provided within the piston rod, an opening of the accommodating passage is sleeved and fixed on the screw nut, a piston which has a diameter as same as an inner diameter of the piston sub-casing is fixed at one end of the piston rod away from the screw nut, the piston divides the sample chamber into a left chamber and a right chamber.

11. The multi-pump integrated mixed fluid delivery device according to claim **10**, wherein an external surface of the opening of the accommodating passage is as same as an inner diameter of the piston sub-casing, an interior of the piston sub-casing is divided into two hydraulic oil chambers.

12. The multi-pump integrated mixed fluid delivery device according to claim **11**, wherein a positioning key is provided on the external surface of the opening, a feather keyway matching the positioning key is provided within the piston sub-casing, the two hydraulic oil chambers are interconnected with each other through the feather keyway.

13. The multi-pump integrated mixed fluid delivery device according to claim **10**, wherein the left chamber and the right chamber are interconnected with a sample input passage which is communicated with an outside world, and a sample output passage which is communicated with the sample cylinder respectively.

14. The multi-pump integrated mixed fluid delivery device according to claim **10**, wherein an isolation sleeve, which is provided in a middle portion of the piston rod, is connected with the piston sub-casing and the sample sub-casing through external threads of the isolation sleeve respectively, one end of the isolation sleeve closer to the screw nut has an inner annular groove, a metal ring which contacts the external surface of the piston rod is provided within the inner annular groove through interference fit, a movable sealing ring is provided within an inner hole which is in contact with the piston rod.

15. The multi-pump integrated mixed fluid delivery device according to claim **2**, wherein the first hydraulic plunger pump and the second hydraulic plunger pump are set within the casing in parallel, the output shaft of the dual-shaft motor is connected with one of the gears through a transmission shaft.

16. The multi-pump integrated mixed fluid delivery device according to claim **15**, wherein the first hydraulic plunger pump and the second hydraulic plunger pump are installed within two chambers of the casing through external threads, the two chambers are interconnected with each other and full of hydraulic oil.

17. The multi-pump integrated mixed fluid delivery device according to claim **1**, wherein:

the multiple hydraulic plunger pumps comprise three hydraulic plunger pumps which are a first hydraulic plunger pump, a second hydraulic plunger pump and a third hydraulic plunger pump;

the first hydraulic plunger pump and the second hydraulic plunger pump are installed in parallel;

a third drive shaft of the third hydraulic plunger pump is opposite to a first drive shaft of the first hydraulic plunger pump;

an output end of the first hydraulic plunger pump, an output end of the second hydraulic plunger pump and an output end of the third hydraulic plunger pump are connected with a respective oil output pipe;

a second gear of the gears is installed on a second drive shaft of the second hydraulic plunger pump;

a first gear of the gears is installed on the first drive shaft of the first hydraulic plunger pump and a third gear of the gears is installed on the third drive shaft of the third hydraulic plunger pump;

the second gear is engaged with the first and third gears.

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